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Arteriosclerosis, Thrombosis, and Vascular Biology

ORIGINAL RESEARCH

Systemic Loss of FKBPL Uncovers Diabetes-Dependent Pathways of Myocardial and Vascular Injury

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BACKGROUND: Impaired angiogenesis underpins cardiovascular disease, particularly in people with diabetes; however, molecular mechanisms are still poorly understood. This study aims to decipher the role of an emerging antiangiogenic protein, FKBPL (FK506-binding protein–like), on cardiac structure and function, vascular integrity, and inflammatory signaling in in vivo and in vitro models of diabetes.

METHODS: In vivo, FKBPL transgenic mice (*fkbpl*$^{+/-}$) were used to examine the metabolic and cardiovascular function and FKBPL-mediated mechanisms in streptozotocin-induced diabetes. In addition, comprehensive in vitro assessments of endothelial function and mechanisms were performed in normal/high-glucose and high/low-FKBPL conditions.

RESULTS: *Fkbpl*$^{+/-}$ mice show signs of early cardiac remodeling (increased E/A ratio, P=0.047, and cardiomyocytes size, P=0.0018; reduced collagen deposition, P=0.013; *col1a1* mRNA reduction, P=0.0028) and aberrant expression of cardiac vascular dysfunction proteins (ICAM-1 [intercellular adhesion molecule 1], P<0.001, SIRT-1 [sirtuin 1]; P<0.001). Proinflammatory cardiac profile was prominent in *fkbpl*$^{+/-}$ murine hearts with increased protein expression of ICAM-1, IL (interleukin)-12p40, IL-15, IL-22, LIF (leukemia inhibitory factor), lipocalin-2, MMP (matrix metalloproteinase)-3/-9, periostin, serpin E1, and VCAM-1 (vascular cell adhesion molecule 1), which were decreased in diabetes. In diabetic mice with low FKBPL expression, glucose metabolism deteriorated, whereas vascular dysfunction improved. In normal glucose conditions, FKBPL knockdown in human aortic endothelial cells reduced VE-cadherin (vascular endothelial cadherin; P=0.0016) and impaired endothelial barrier (P<0.001). In high-glucose conditions, endothelial FKBPL knockdown improved angiogenesis, however overexpression of FKBPL reduced angiogenesis by inhibiting the FGF (fibroblast growth factor) 2 and PDGF (platelet-derived growth factor) pathways (P<0.001) and increasing proinflammatory pathways (TGF [transforming growth factor]-β, P<0.001; leukocyte migration, P=0.033; IL-7 signaling, P=0.039), by upregulating miR-29b-3p (P=0.01) and miR-302b-5p (P=0.03), likely via CD44. FKBPL-based peptide mimetic, AD-01 (1 nM), in high-glucose conditions, upregulated endothelial *vcam1* and *glut1* mRNA expression, independent of miR-302b-5p.

CONCLUSIONS: FKBPL plays an important role in glucose metabolism, endothelial function, angiogenesis, cardiac inflammation and function, and could be explored as a therapeutic target of cardiovascular disease both in nondiabetes and diabetes settings using precision medicine approach.

Key Words: angiogenesis ■ cardiovascular diseases ■ endothelial cells ■ mice ■ streptozocin

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Cardiovascular disease (CVD) is the leading cause of mortality worldwide, with diabetes increasing the risk of CVD 2- to 4-fold in adults.[1] Atherosclerosis is the main cause of CVD, underpinned by the narrowing of the arteries, obstructing the supply of oxygen and nutrients to the heart.[2] Healthy vascular endothelial cells that line the blood vessels control the permeability, preventing the influx of damaging substances. The hyperglycemia

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Nonstandard Abbreviations and Acronyms

Ang-II	angiotensin II
CVD	cardiovascular disease
DAPI	4′,6-diamidino-2-phenylindole
ECM	extracellular matrix
eNOS	endothelial NO synthase
FKBPL	FK506 binding protein–like
HAoEC	human aortic endothelial cell
HCAEC	human coronary artery endothelial cell
ICAM	intercellular adhesion molecule
IL	interleukin
JCAD	junctional cadherin–associated-5
JNK	C-jun N-terminal kinase
LV	left ventricle
MAPK	mitogen-activated protein kinase
MMP	matrix metalloproteinase
NF-κB	nuclear factor-κB
PDGF	platelet-derived growth factor
RBP4	retinol-binding protein 4
SIRT-1	sirtuin 1
TGF	transforming growth factor
TLR	toll-like receptor
TNF-α	tumor necrosis factor-α
VCAM	vascular cell adhesion molecule
VE-cadherin	vascular endothelial cadherin
VEGF	vascular endothelial growth factor
WT	wild type

What Are the Clinical Implications?

This study identifies FKBPL (FK506-binding protein–like) as a context-dependent regulator of cardiovascular pathology with direct clinical implications for diabetes-associated heart disease. In diabetic settings, reduced baseline FKBPL expression exerts a cardioprotective effect by attenuating cardiac inflammation and promoting angiogenesis, whereas in nondiabetic conditions, low FKBPL levels are associated with endothelial dysfunction, aberrant angiogenesis, and early cardiac remodeling and inflammation. We further identify distinct downstream FKBPL signaling pathways under normoglycemic versus hyperglycemic conditions, highlighting the importance of metabolic context in therapeutic targeting. Notably, the FKBPL-based peptide therapeutic, AD-01, failed to confer endothelial protection under hyperglycemic stress, despite prior evidence of stress-induced negative feedback regulation of FKBPL. Collectively, these findings position FKBPL as a clinically relevant, stratified therapeutic target: suppression of FKBPL may be beneficial in patients with diabetes to limit cardiovascular inflammation and injury, whereas restoration of physiological FKBPL levels may be required to preserve vascular function in nondiabetic populations. These data underscore the need for metabolic context–specific FKBPL-directed therapies and caution against uniform application of FKBPL-based interventions across patient groups.

in diabetes triggers proinflammatory signals that prompt endothelial cells to produce adhesion molecules, affecting the integrity of junction proteins, thus increasing permeability. This allows leukocytes to penetrate the arterial wall, initiating inflammatory processes that contribute to atherosclerosis.[3,4]

A fundamental role of normal vascular endothelium is managing angiogenesis, crucial for tissue development and wound healing. This process requires a tightly controlled balance between proangiogenic and antiangiogenic factors to stimulate vessel growth and remodel existing networks.[5] However, under stress conditions, including hyperglycemia, this equilibrium is disrupted, leading to vascular damage.[6] In cardiac tissue, hyperglycemia results in restricted angiogenesis, reducing the heart's ability to generate new blood vessels and maintain normal function.[7,8]

FKBPL (FK506-binding protein–like), a divergent member of the immunophilin protein family, is a potent secreted antiangiogenic protein.[9,10] FKBPL in complex with HSP90 and steroid receptors plays a key role in regulating the signaling of glucocorticoid receptors, relevant to metabolism.[11,12] In people with type 2 diabetes or CVD, plasma FKBPL concentrations are increased and associated with age, gender, total cholesterol, and systolic blood pressure, key determinants of CVD.[13] Various studies have shown that FKBPL is critical in regulating developmental, physiological, and pathological angiogenesis.[12,14,15] For example, the absence of FKBPL led to embryonic lethality in mice, underlining its pivotal role in embryonic development, whereas mice with heterozygous knockdown of FKBPL developed normally, but their blood vessels exhibited structural leakage and compromised integrity,[15] providing evidence for FKBPL's crucial role in healthy vasculature. Despite these findings, FKBPL's role in the pathogenesis of CVD in the presence or absence of diabetes remains unexplored.

Although previous work has shown that FKBPL plasma concentration is increased in people with type 2 diabetes, CVD,[13] or HFpEF,[16] its role and mechanisms in these conditions are unknown, particularly in terms of how it regulates cardiac function and structure, vascular integrity, endothelial function, and metabolic stress. To examine whether FKBPL haploinsufficiency *(fkbpl*$^{+/-}$*)* affects cardiac remodeling, endothelial homeostasis, and the inflammatory response within a setting of

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streptozotocin-induced diabetes, we designed an in vivo study to address these phenotypes. We also specifically investigated the impact and mechanism on endothelial cells of high and low FKBPL expression in normal and high-glucose conditions to identify downstream mechanisms.

In this study, we provide the first comprehensive in vivo evidence that FKBPL regulates glucose metabolism, cardiac inflammation, and endothelial function, in the absence and presence of diabetes. We show that adult *fkbpl*$^{+/-}$ transgenic mice have a proinflammatory cardiac profile, which is abrogated in diabetes. These findings are complemented by in vitro work using human aortic endothelial cells (HAoECs). Overall, our study shows that low FKBPL expression has a protective role in cardiovascular function in the presence of diabetes, whereas it could be damaging to vascular integrity in the absence of diabetes. Targeting FKBPL unveils a novel personalized therapeutic strategy for CVD patients, both in the presence and absence of diabetes.

MATERIALS AND METHODS

Further details on methodology are available in the Supplemental Material. The data supporting the findings are available from the corresponding author on reasonable request.

Experimental Animals

All in vivo experiments were performed in accordance with the UK Animals (Scientific Procedures) Act 1986 and approved by the Queen's University Belfast Animal Welfare and Ethical Review Body. Mice were housed in groups of up to 5, under standard conditions of a 12-hour light-dark cycle and had ad libitum access to water and standard chow (Teklad Global 18% Protein Rodent Diet, Inotiv). Nondiabetic mice were maintained at a constant temperature of 21 °C, whereas diabetic mice were maintained at 28 °C to prevent hypothermia and temperature rhythm disruption.[17] All mice used in experiments were male wild-type (WT) C57BL/6J mice or FKBPL transgenic mice (*fkbpl*$^{+/-}$) bred from an established colony at Queen's University Belfast,[15] aged between 8 and 12 weeks, with a body weight of >20 g. Animals were allowed to acclimatize for 7 days before experimental procedures.

Streptozotocin-Induced Diabetes in Mice

Mice were fasted for 4 hours before the experiment. Streptozotocin (Sigma-Aldrich, United Kingdom) was dissolved in 0.1 mol/L citrate buffer (pH 4.5) at 10 mg/mL and injected intraperitoneally at 50 mg/kg for 5 consecutive days. Control mice received a citrate buffer injection. Mice were given 10% sucrose solution in their drinking water to prevent hypoglycemia. Diabetes was confirmed if blood glucose levels exceeded 15 mmol/L, 10 days after the final injection. Glucose levels were monitored using tail vein puncture weekly for 12 weeks using a glucometer. At 12 weeks, echocardiography and insulin tolerance tests were performed. At 13 weeks, mice were euthanized, and organs were collected for further analysis.

Transthoracic Echocardiography

Echocardiographic assessment of the left ventricle (LV) was conducted on lightly anesthetized mice using 2.5% isoflurane/oxygen. The LV parameters were quantified through short axis scans at the papillary muscle level, employing a high-frequency 45 MHz transducer attached to a Vevo 770 imaging system (VisualSonics, Canada). One-dimensional M-mode scans were recorded for LV measurements, whereas pulse-wave Doppler imaging was used to assess the mitral valve E/A ratio as an indicator of diastolic dysfunction. Three measurements were taken for each animal using Vevo 770 v3.0.0 software (VisualSonics).

Insulin Tolerance Test Assessment

All mice were fasted for 4 hours before insulin injection. Actrapid insulin (100 U/mL; AAH Pharmaceutics, United Kingdom) was dissolved in 0.9% sodium chloride. Each mouse received 1 U/kg IP, followed by blood glucose measurements via tail vein puncture at 0, 15, 30, 60, and 120 minutes later by glucometer (Accu Check Performa, Roche, United Kingdom).

Tissue Collection and Processing

Mice were euthanized with intraperitoneal injections of ketamine (100 mg/kg) and xylazine (10 mg/kg). Blood was collected by cardiac puncture into EDTA-coated tubes, and HbA1c was measured immediately using the A1C now kit (BHR Pharmaceuticals Ltd, United Kingdom) according to the manufacturer's instructions. Hearts were removed, and the LV was divided longitudinally into 4 equal pieces from base to apex, following the long axis of the heart. One piece was fixed for morphological analysis, another for immunofluorescence staining. The remaining 2 pieces were snap-frozen and stored at −80 °C for gene expression studies.

Immunofluorescence Staining

LV samples were fixed in 4% paraformaldehyde for 24 hours at 4 °C and cryoprotected in sucrose (Sigma-Aldrich, United Kingdom) gradients (10%, 20%, and 30%). They were then embedded in O.C.T (Tissue-Tek, the Netherlands) and snap-frozen on dry ice, stored at −20 °C for later use. Sections of 10 µm thickness were cut using a cryostat (Leica CM1900, Germany) at −20 °C and placed on superfrost slides (ThermoScientific, United Kingdom). After thawing, sections were washed with PBS followed by incubation with primary antibodies: FKBPL (1:150; Proteintech, United Kingdom), SIRT-1 (sirtuin 1; 1:150; Abcam, United Kingdom), ICAM-1 (intercellular adhesion molecule 1; 1:200; ThermoScientific, United Kingdom), and CD-31 (1:200; R&D systems), diluted in PBS with blocking agents and overnight incubation at 4 °C. Sections were then washed and secondary antibodies, Alexaflour affinipure (1:250; Jackson Immunoresearch, United Kingdom) and Cy3 affinipure (1:250; Jackson Immunoresearch, United Kingdom) added and incubated at room temperature for 1 hour in the dark. After further PBS washes, slides were mounted with DAPI (4′,6-diamidino-2-phenylindole)-containing mounting media (Vectashelid H-1200, Vector Laboratories, United Kingdom) and coverslips. Negative staining controls were included by omitting the primary antibody and incubating sections with secondary antibody only. No specific staining was observed in these controls.

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Images were taken at ×200 magnification using a Leica diabetesi8 fluorescent microscope, and image analysis was performed using ImageJ software by quantifying the intensity of the selected tissue area and subtracting background intensity. Values were normalized to DAPI intensity.

Histology

Paraformaldehyde-fixed LV tissues were dehydrated using a wax tissue processor (Leica TP 1020, Germany) using ethanol and processed for hematoxylin and eosin or picrosirius red staining.[18] The stained slides were examined at ×40 magnification using a Nikon Eclipse 80i microscope (Nikon). Cardiomyocyte area (μm^2) was measured in 5 random LV sections using hematoxylin and eosin–stained slides, analyzing 2 separate fields of view in each section. Fifty cells per LV sample were analyzed with a single, centrally located nucleus, and data quantified by blinded digital image analysis (NIS-Elements, Nikon); a mean value for each sample was generated.

Mouse Cardiac Cytokine Proteome Array

The Proteome Profiler Mouse XL Cytokine array kit (R&D Systems, United Kingdom) was used to assess cardiac cytokine expression, following the manufacturer's instructions. Briefly, pooled LV tissues per group were homogenized, centrifuged, and protein quantified using the BCA assay. Proteome membranes were incubated in blocking buffer for 1 hour, followed by overnight incubation at 4 °C. The next day, membranes were incubated with primary and secondary antibody cocktails. After washing, membranes were imaged using the G-box, and images were analyzed using HLimage++ software (Western Vision Software). The data were presented as a heat map generated by Microsoft Excel.

Cell Culture

HAoECs (Lonza, United Kingdom) or human coronary artery endothelial cells (HCAECs; ATCC) were cultured in complete EBM-2 medium containing 5.5 mmol/L glucose and 2% fetal bovine serum (Lonza, United Kingdom). Cells were maintained at 37 °C with 5% CO_2, passaged at 80% confluence using trypsin-EDTA 0.05%, and cultured in collagen-coated flasks or plates. High-glucose media was formulated by adjusting D-glucose levels to a final concentration of 30 mmol/L. Mannitol was used as an osmotic control, incorporating 24.5 mmol/L D-glucose for the high-glucose treatment and 24.5 mmol/L mannitol for the osmotic control condition. Endothelial cells were not used beyond passage 9.

Tubule Formation Assay

Tubule formation assay was performed as described.[19] Quantification of branches and junction formation was assessed in 6 distinct fields of view using the ImageJ angiogenesis analyzer plugin.

FKBPL Transfection Using Overexpression Plasmid

HAoECs (700 000) were seeded overnight in 60-mm cell culture dishes. The following day, the transfection solution was prepared: 1 µg plasmid (pFKBPL or pCMV3; Sino Biological), 3 µL Endofectin (GeneCopoeia), and 50 µL reduced-serum OptiMEM media (Thermo Fisher, United Kingdom). After a 10-minute incubation at room temperature, media was removed, and the cells were washed with reduced-serum OptiMEM before OptiMEM supplemented with 2% fetal bovine serum. Cells were incubated overnight at 37 °C with 5% CO_2. The next day, the medium was replaced with fresh culture media for 24 hours.

FKBPL Knockdown Using siRNA

ON-Target plus FKBPL siRNA and nontargeting siRNA were reconstituted to 20 µmol/L concentrations in RNAse-free water. HAoECs were seeded overnight, 350 000 per well in 6-well plates before the addition of transfection medium containing 25 nM FKBPL or nontargeting siRNA combined with Dharmafect (Dharmacon/Horizon Discovery, United Kingdom). Cells were harvested for experiments 72 hours later.

Endothelial Cell Barrier Function Assay

Endothelial cell barrier formation was assessed using the xCELLigence Real-Time Cell Analysis Instrument (ACEA Bioscience). Further information is available in the Supplemental Methods.

MicroRNA Sequencing

FKBPL was upregulated in HAoECs for 24 hours before treating with high glucose (30 mmol/L) or low glucose (5.5 mmol/L) for an additional 24 hours. MicroRNAs were extracted using the miRNeasy mini kit (QIAGEN, United Kingdom) following the manufacturer's instructions. Further information is available in the Supplemental Methods. MicroRNA data have been loaded into the ArrayExpress collection in BioStudies with accession number: E-MTAB-16356.

AD-01 Treatment and MicroRNA Transfection of HCAECs

HCAECs (ATCC) were maintained in complete EBM-2 medium and 2% fetal bovine serum (Lonza, United Kingdom) at 37 °C with 5% CO_2. Cells were seeded into 6-well plates at a density of 9×105 cells per well in 1 mL antibiotic-free medium. The following day, cells were treated with AD-01 to a final concentration of 1 nM or 100 nmol/L. After 24 hours, the media was replaced with 5.5 and 30 mmol/L D-glucose-containing medium and AD-01 treatment. For microRNA transfection, miR-302b-5p inhibitor (1 µmol/L stock; Shanghai GenePharma) was diluted to a final working concentration of 20 nmol/L using Lipofectamine RNAiMAX (Invitrogen), following the standard RNAiMAX complex-formation protocol. Transfection complexes were added directly to each well together with AD-01 (1 nM), and cells were incubated for 24 hours at 37 °C and 5% CO_2 before downstream analysis.

Real-Time Polymerase Chain Reaction

Total RNA was isolated from tissues using the Trizole-chloroform extraction method and complementary cDNA synthesized.[18] RNA was extracted from HCAECs using TRI-reagent and cDNA generated using the iScript cDNA synthesis kit (Australia; PMID: 39365855; Bio-Rad). Real-time polymerase chain reaction was conducted using a Lightcycler 480 II (Roche, Switzerland) or a CFX96 Thermocycler (Bio-Rad, Australia)

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using iQ SYBR (Bio-Rad, Australia). *Fkpbl* and *β-actin* Taqman monohydrolyzed primer probes obtained from Roche were used for real-time polymerase chain reaction. Relative mRNA expression changes were analyzed using the ΔΔCt method, with normalization to β-actin. Primer sequences for mouse genes use are as follows: *postn* (F: CCATTAACGGAATCAAGAATGG; R: AACTTGTTTGGCAGAATCAG), *fsp-1* (F: TATTCAGCACTTCCTCTCTC; R: CCTCTTTGCCTGAGTATTTG), *col1a1* (F: CGTATCACCAAACTCAGAAG; R: GAAGCAAA GTTTCCTCCAAG), *col3a1* (F: ACTCAAGAGTGGAGAATACTG; R: AACATGTTTCTTCTCTGCAC), *ctfg* (F: GAGGAAAA CATTAAGAAGGGC; R: AGAAAGCTCAAACTTGACAG), *fn-1* (F: CCTATAGGATTGGAGACACG; R: GTTGGTAAATAGCTGTTC GG). Primer sequences for human genes used are as follows: *glut1* (F: TTCACTGTCGTGTCGCTGTTTG; R: TCACAC TTGGGAATCAGCCCC), *icam-1* (F: GTATGAACTGAGC AATGTGCAAG; R: GTTCCACCCGTTCTGGAGTC), *vcam-1* (F: TTTGACAGGCTGGAGATAGACT; R: TCAATGTGTAATTTAGC TCGGCA), *ir* (F: AAAACGAGGCCCGAAGATTTC; R: GAGC CCATAGACCCGGAAG).

Western Blotting

Protein lysates from frozen tissue samples (30 mg) or HCAECs were extracted using RIPA buffer supplemented with protease inhibitor cocktail and subjected to Western blotting. Membranes were blocked with 5% nonfat milk and then incubated with primary antibodies, FKBPL (1:1000; Proteintech, United Kingdom), VE-cadherin (vascular endothelial cadherin; 1:500; Abcam, United Kingdom), overnight at 4°C on a shaker. After washing, membranes were incubated with species-specific HRP-conjugated secondary antibody (diluted 1: 2000 in 5% milk; Cell Signaling) for 1 hour at room temperature. Chemiluminescent bands were detected using West Femto Maximum sensitivity substrate (Thermo Fisher, United Kingdom) and visualized with the G-box (Kodak, United Kingdom). Blots were processed sequentially on the same membrane without cutting or stripping. Target proteins were detected and imaged first, followed by probing for the loading control and reimaging. Protein expression was quantified using ImageJ software; band intensities were normalized against the loading control, GAPDH (1:1000; Cell Signaling).

Statistical Analysis

All data were presented as mean±SD or SEM derived from n animals, tissues, or cell-based experimental replicates. Cell-based experiments involved a minimum of 3 biological replicates. Statistical analyses were performed using GraphPad Prism 5.0 (GraphPad) using unpaired Student *t* tests, or 1-way or 2-way ANOVA, followed by Sidak or Bonferroni post hoc multiple comparison tests, for normalized data. Shapiro-Wilks test was used to check whether the data are normally distributed. A $P<0.05$ was considered significant.

RESULTS

Low FKBPL Expression Leads to Early Cardiac Remodeling in Mice

FKBPL mRNA and protein expression were significantly reduced in *fkbpl*$^{+/-}$ nondiabetic mice compared with WT controls. Diabetes induction using streptozotocin injection decreased FKBPL expression in WT mice compared with nondiabetic WT controls (Figure 1A and 1B; n≥5; mRNA: $P=0.016$, protein: $P<0.001$). Notably, diabetic *fkbpl*$^{+/-}$ mice exhibited significantly lower cardiac FKBPL expression compared with diabetic WT mice (Figure 1A and 1B, n≥5; mRNA: $P=0.0049$, protein: $P<0.001$). FKBPL suppression resulted in reduced cardiac collagen content in nondiabetic mice (Figure 1C, n≥7; $P=0.013$) but increased cardiomyocyte area (Figure 1D, n≥7; $P=0.0018$). These findings imply that basal FKBPL suppression in mice leads to early cardiac remodeling. Interestingly, where both diabetic groups increased collagen ratio (Figure 1C; $P<0.001$), cardiomyocyte area was only reduced between nondiabetic and diabetic *fkbpl*$^{+/-}$ mice (Figure 1D; $P<0.001$).

To determine if changes in FKBPL expression impact cardiac function, noninvasive transthoracic echocardiography was performed (Table 1). Pulsed-wave Doppler measurements revealed increased mitral valve E/A ratio in nondiabetic *fkbpl*$^{+/-}$ when compared with WT controls (Table 1, n≥8; $P=0.047$); in both diabetic groups, E/A ratio was significantly reduced compared with their genotype controls (WT versus WT streptozotocin, $P=0.0039$; *fkbpl*$^{+/-}$ vehicle control versus *fkbpl*$^{+/-}$ streptozotocin; $P<0.001$). There were no significant differences in systolic parameters: fractional shortening ($P=0.21$), ejection time ($P=0.21$), LV end-systolic diameter ($P=0.15$), or volume ($P=0.23$) between WT control and WT diabetic mice. Similarly, no significant differences were observed between WT diabetic and *fkbpl*$^{+/-}$ diabetic mice. These data indicate that at this stage (12 weeks), diabetes induction did not appreciably change systolic function in WT mice, and that decreased expression of FKBPL did not further worsen systolic dysfunction in diabetic animals.

Diabetes Protects Against *fkbpl*$^{+/-}$-Induced Cardiac Remodeling and Inflammation

Mice were rendered diabetic, and blood glucose levels were monitored weekly for 12 weeks after streptozotocin injection. As expected, blood glucose was significantly higher in diabetes WT and *fkbpl*$^{+/-}$ mice when compared with nondiabetic controls (Figure 2A and 2B; $P<0.001$), but diabetes *fkbpl*$^{+/-}$ had ≈15% higher blood glucose levels than diabetes WT mice (Figure 2B; $P<0.001$). No differences were observed between genotypes under nondiabetic conditions (Figure 2A and 2B); however under diabetic conditions, aligned to blood glucose findings, low FKBPL led to modestly higher HbA1c than diabetes WT mice (Figure 2C; $P=0.014$). As our mice were only exposed to diabetes for 12 weeks, it is likely that this difference in HbA1c will continue to increase in magnitude with longer duration of diabetes. Aligned to this, previous work using human plasma samples showed that FKBPL was negatively correlated with fasting blood glucose and HbA1c in people with type 2 diabetes, but

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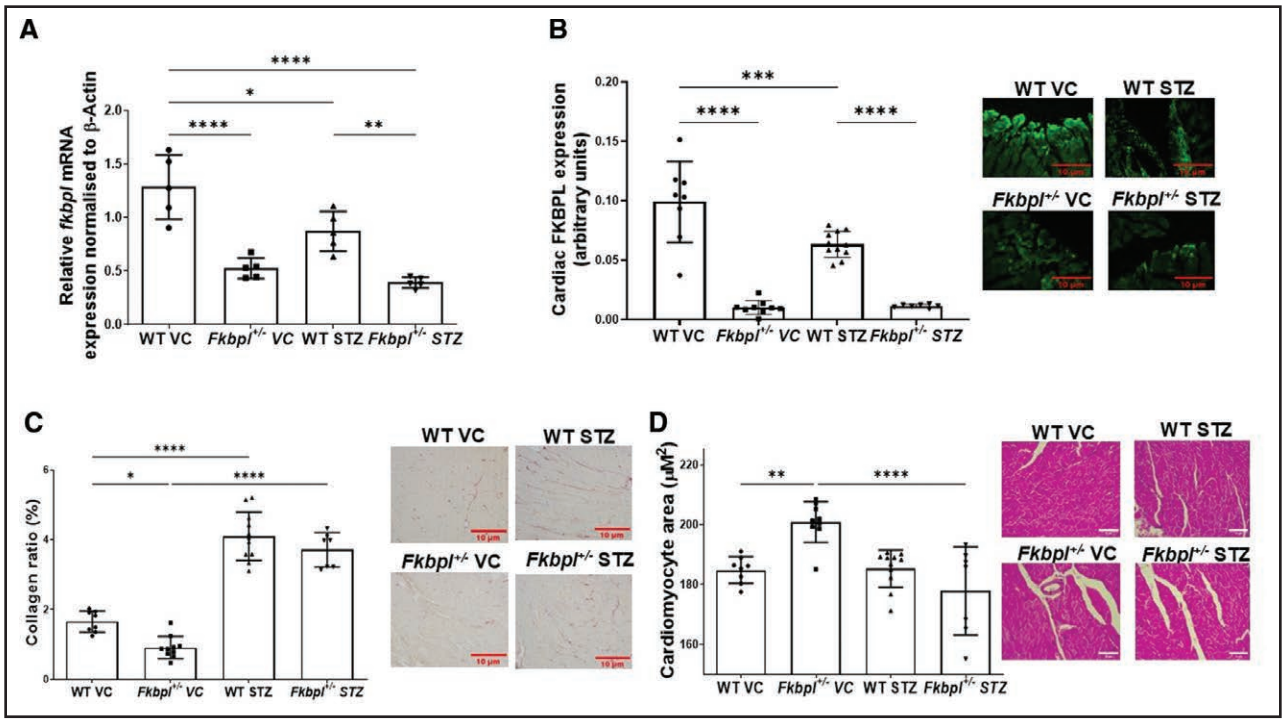

Figure 1. Low background FKBPL (FK506-binding protein–like) expression reduces collagen content and increases cardiomyocyte size in nondiabetic mice.

A, RNA was extracted from murine heart tissues, and FKBPL mRNA expression was determined in nondiabetic vehicle control (VC) and diabetic streptozotocin (STZ) treated wild-type (WT) and *fkbpl*$^{+/-}$ mice (n=5, 1-way ANOVA, Sidak post hoc test, **P*<0.05, ***P*<0.01, ****P*<0.001). **B**, Frozen left ventricle (LV) heart sections were processed and stained before FKBPL protein expression was quantified using mean fluorescence and compared between the 4 groups; representative immunofluorescence images of FKBPL (green) inset. **C**, LV sections were stained with picosirius red, and collagen deposition was quantified and compared between the 4 groups of mice. **D**, Cardiomyocyte area was assessed in LV sections after staining with hematoxylin and eosin, and the cross-sectional area of cardiomyocytes was compared between the 4 groups of mice (mean±SD; n≥7 [biological replicates]; 1-way ANOVA with Sidak post hoc test; **P*<0.05, ***P*<0.01, ****P*<0.001, *****P*<0.0001).

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positively correlated with the duration of type 2 diabetes.[13] to assess the effect of FKBPL on insulin signaling, an insulin tolerance test was performed at 12 weeks of diabetes induction. No difference in blood glucose levels was observed between WT and *fkbpl*$^{+/-}$ mice with (*P*=0.298) or without diabetes (*P*=0.99; Figure 2D and 2E, n≥7).

We next examined the cardiac inflammatory profile using a mouse proteome profiler XL cytokine assay (Figure 3A). Several inflammatory markers were altered with FKBPL suppression under basal conditions when compared with WT mice. For example, coagulation factor III, cystatin C, ICAM-1, IL (interleukin)-12p40, IL-15, IL-22, LIF (leukemia inhibitory factor), lipocalin-2, MMP (matrix metalloproteinase) 3, MMP-9, periostin, RAGE, RBP4 (retinol-binding protein 4), Reg3G, Serpin E1, and VCAM-1 (vascular cell adhesion molecule 1) were all upregulated, representing proinflammatory factors associated with cardiac inflammation, endothelial dysfunction and remodeling[20–23] (Figure 3B). In contrast, ICAM-1, IL-12p40, IL-15, IL-22, LIF, Lipocalin-2, MMP-3, MMP-9, periostin, serpin E1 and VCAM-1 were reduced in diabetes *fkbpl*$^{+/-}$ mice. Interestingly, adiponectin, an adipocyte-derived hormone capable of reversing insulin resistance and associated with obesity[24] was also reduced in diabetes *fkbpl*$^{+/-}$ but markedly increased in *fkbpl*$^{+/-}$ (Figure 3B). FGF (fibroblast growth factor)-21, Flt-3 ligand, Gas 6, IGFBP5, IGFBP6, IL-1α, IL-1β, IL-11, leptin, pentraxin 3, and E-selectin were markedly reduced in diabetes *fkbpl*$^{+/-}$ mice; most of these proteins were also decreased or unchanged in nondiabetic *fkbpl*$^{+/-}$ mice compared with WT controls (Figure 3B). These findings imply that while diabetes impairs diastolic function and increases fibrosis, these changes may not be related to diabetes-induced FKBPL suppression. Overall, myocardial inflammatory cytokine expression was high in diabetes WT and in nondiabetic *Fkbpl*$^{+/-}$ mice, but reduced in *Fkbpl*$^{+/-}$ mice with diabetes.

Low Baseline Levels of FKBPL May Impair Cardiac Vascular Function in the Absence of Diabetes but May Be Protective in the Presence of Diabetes

Previous studies reported that vascular dysfunction in *fkbpl*$^{+/-}$ mice was associated with increased

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Table 1. Echocardiographic Data at Week 12 of Diabetes in STZ-Induced Diabetic WT and Fkbpl$^{+/-}$ Mice

Measurements	VC		STZ		*P* value
	WT (n=8)	*Fkbpl*$^{+/-}$ (n=9)	WT (n=11)	*Fkbpl*$^{+/-}$ (n=7)	
E/A ratio	1.7±0.06	2±0.1*	1.3±0.06†	1.3±0.05‡	<0.001§
Fractional shortening, %	29±1.9	25.1±1.1	33.7±1.9‖	29±1.7	0.0061§
Isovolumetric contraction time, ms	16.7±1.5	15.8±0.9	15.9±0.8	14.6±0.9	0.64
Ejection time, ms	56.0±3.6	55.8±3.2	53.3±2.7	52.5±1.6	0.80
IVRT, ms	20.1±1.6	17.6±0.9	23.5±1.4‖	19.0±0.8	0.008§
MPI	0.68±0.07	0.61±0.02	0.80±0.06	0.60±0.03	0.12
LV posterior wall diastole, mm	1.2±0.1	1.3±0.08	1.2±0.08	1.2±0.1	0.93
LV posterior wall systole, mm	0.82±0.06	0.90±0.07	0.80±0.1	0.90±0.1	0.79
LV posterior wall thickness, %	49.5±3.4	47.3±6.7	56.3±14.7	46.8±9.8	0.90
LV end-systole diameter, mm	2.7±0.1	2.9±0.1	2.4±0.1‖	2.7±0.2	0.019§
LV end-diastole volume, μL	64.2±3.6	64.6±3.7	53.7±3.5	59.6±5.5	0.16
LV end-systole volume, μL	28.3±2.5	32.4±2.5	20.5±2.3‖	26.8±4.2	0.029§
LV ejection fraction, %	56.0±2.9	50.1±1.9	63.0±2.6‖	56.3±2.6	0.0059§

Data are presented as mean±SEM. One-way ANOVA followed by Sidak multiple comparison test was used for statistical comparison. IVRT indicates isovolumetric relaxation time; LV, left ventricular; MPI, myocardial performance index; STZ, streptozotocin; VC, vehicle control; and WT, wild-type.

*Significant difference between nondiabetic fkbpl$^{+/-}$ compared with nondiabetic WT.

†Significant difference between nondiabetic WT compared with diabetic WT.

‡Significant difference between nondiabetic fkbpl$^{+/-}$ compared with diabetic fkbpl$^{+/-}$.

§

‖Significant difference between nondiabetic fkbpl$^{+/-}$ compared with diabetic WT.

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angiogenesis,[15] so we examined cardiac angiogenic capacity in mice with and without diabetes. As expected, CD-31 expression levels were significantly upregulated in both diabetes and nondiabetic *fkbpl*$^{+/-}$ mice compared with their respective WT controls (Figure 4A, n≥7; *P*<0.001). Increased ICAM-1 levels are an early marker of endothelial dysfunction,[25] and its expression was significantly upregulated in nondiabetic *fkbpl*$^{+/-}$ versus WT mice (Figure 4B, n≥7; *P*<0.001). ICAM-1 expression was also induced with diabetes in WT mice (Figure 4B, n≥7; *P*<0.001), but lower in diabetic *fkbpl*$^{+/-}$ versus WT diabetes mice (Figure 4B, n≥7; *P*=0.016), validating our findings from Figure 3 in terms of reduced inflammation in diabetic conditions. SIRT-1 plays a key role in improving endothelial function in diabetes[26]; SIRT-1 protein expression was assessed in murine hearts and found to be downregulated in *fkbpl*$^{+/-}$ and diabetic WT mice, when compared with their WT controls (Figure 4C, n≥7; *P*<0.001, *P*=0.004). Interestingly, SIRT-1 expression was restored in *fkbpl*$^{+/-}$ mice in the presence of diabetes compared with diabetic WT mice (Figure 4C, n≥7; *P*=0.0013). Furthermore, no difference was observed in cardiac VE-cadherin expression between any of the groups (Figure 4D), which could be due to low abundance of endothelial cells within the cardiac tissue. These findings suggest that reduced cardiac FKBPL expression in diabetes may serve as a protective mechanism in the vasculature.

Low Levels of FKBPL Suppress the Expression of *col1a1* in Both Diabetic and Nondiabetic Mice

Given that cardiac inflammation and remodeling were present in both *fkbpl*$^{+/-}$ mice and diabetic WT mice, whereas it was reduced in diabetes *fkbpl*$^{+/-}$ mice, the presence of cardiac fibrosis was investigated. Several cardiac fibrosis genes were assessed including, *postn*, *fsp-1*, *col1a1*, *col3a1*, *ctgf*, and *fn-1*, as previously described.[27] No significant changes were observed in most cardiac fibrosis markers as a result of a different genotype (Figure 5A through 5F, n≥3), except for *col1a1*, which was significantly reduced in both nondiabetic and diabetic *fkbpl*$^{+/-}$*mice* compared with their WT controls (Figure 5C, n≥6; *P*=0.0028, *P*=0.0044), respectively. Two fibrotic genes, *fsp-1* was increased in diabetic versus nondiabetic mice irrespective of genotype (Figure 5B, n≥5; *P*<0.001), whereas *ctgf* was only increased in WT diabetic versus WT nondiabetic mice (Figure E, n≥5; *P*<0.001) with borderline difference between nondiabetic and diabetic *fkbpl*$^{+/-}$ mice (Figure E, n≥5; *P*=0.055).

Silencing FKBPL Impairs Endothelial Cell Barrier Function

Next, we investigated the function of FKBPL in regulating endothelial cell barrier formation using the

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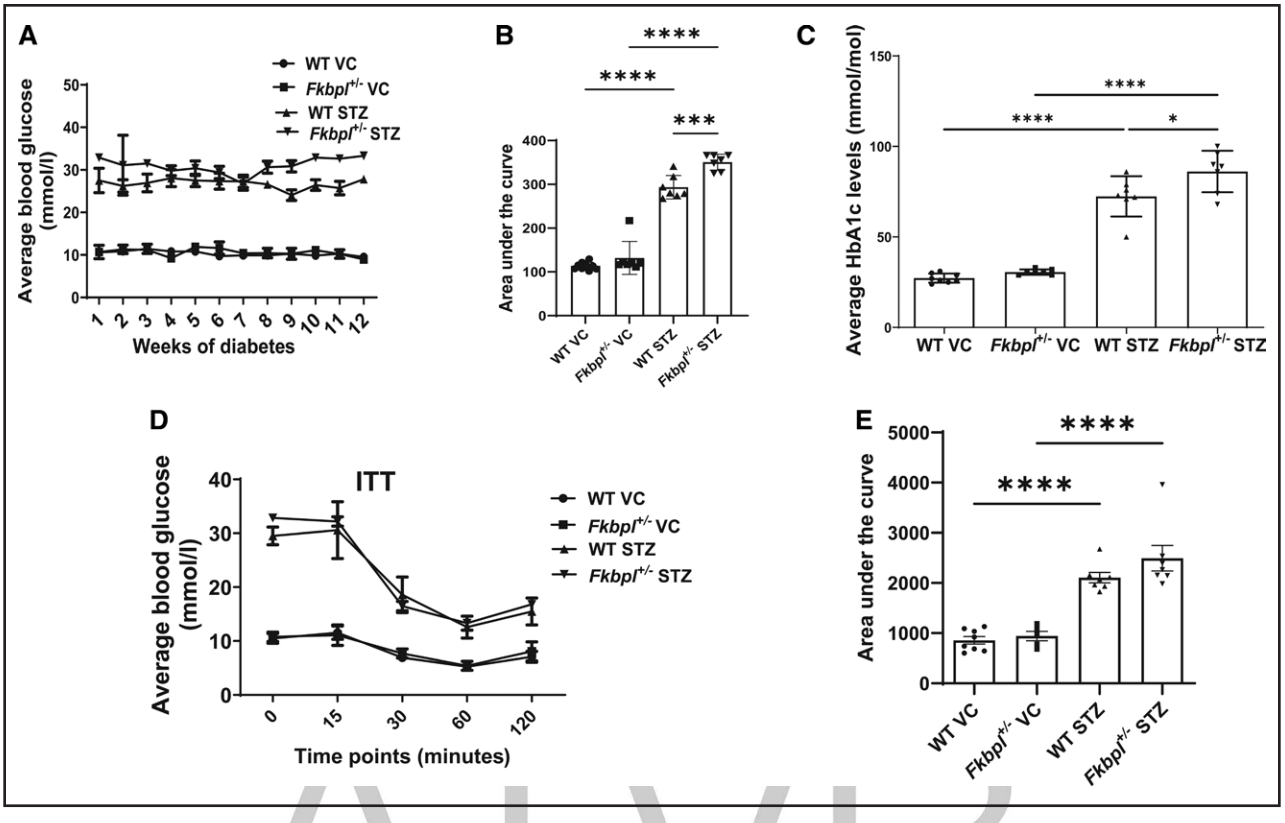

Figure 2. Reduced FKBPL (FK506-binding protein–like) potentiates hyperglycemia in diabetic mice without altering insulin sensitivity.

A, Weekly blood glucose levels over 12 weeks were measured in both diabetic and nondiabetic wild-type (WT) and *fkbpl*$^{+/-}$ mice and compared between the 4 groups. **B**, Area under the curve (AUC) of longitudinal blood glucose was calculated per mouse and grouped, confirming higher AUC in both groups of diabetic mice and diabetic *fkbpl*$^{+/-}$ mice compared with diabetic WT mice. **C**, HbA1c concentration was determined in the blood at the end of the experiment (week 13) and compared between the 4 groups, showing higher HbA1C concentration in both diabetic groups and diabetic *fkbpl*$^{+/-}$ mice compared with WT diabetic mice. **D**, Insulin tolerance testing (ITT) was performed, and (**E**) AUC of blood glucose concentration was calculated at week 12, showing significant increase in both diabetic groups (mean±SD; n≥7 [biological replicate], 1-way ANOVA with Sidak post hoc test, *P<0.05, ***P<0.001, ****P<0.0001). STZ indicates streptozotocin; and VC, vehicle control.

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xCELLigence system. Silencing FKBPL in HAoECs using siRNA technology reduced FKBPL protein expression substantially (Figure 6A, n=3; P=0.0095). FKBPL reduction was associated with impaired barrier function at 8 hours (P<0.001) and 22 hours (P<0.001; Figure 6B through 6D, n=3) when compared with the control siRNA. To assess the permeability of endothelial cells in response to stress, the endothelial cell barrier was disrupted by treating cells with 50 ng/mL recombinant human VEGF (vascular endothelial growth factor) or 0.1% bovine albumin serum as a vehicle control. Endothelial barrier disruption occurred ≈1 to 2 hours after treatment in both groups, indicated by the lowest normalized cell index value (Figure 6E). As expected, VEGF treatment reduced the cell index value in both FKBPL knockdown (P=0.0029) and in control conditions (P<0.001; Figure 6F and 6G, n=3; P<0.001). However, the endothelial cell barrier response was significantly worse when FKBPL was silenced (Figure 6H, n=3, P<0.001). In conjunction with impaired barrier function as a result of FKBPL knockdown, VE-cadherin expression was also reduced (Figure 6I, n=3; P=0.0016).

FKBPL Has a Critical Role in Regulating Endothelial Cell Angiogenic Potential in Hyperglycemia

We next assessed the impact of FKBPL suppression or overexpression on the angiogenic potential of endothelial cells under hyperglycemic conditions. High glucose impaired the angiogenic potential of HAoECs treated with control siRNA (Figure 7A through 7C; P=0.024), but as expected, silencing FKBPL increased the number of junctions and branches under normal and high-glucose conditions (Figure 7A through 7C, n=3, normal glucose: junctions, P=0.0044, branches, P=0.0045; high glucose: P<0.001). On the contrary, FKBPL overexpression suppressed the number of junctions and branches under normal glucose conditions when compared with the empty vector control (junctions: P=0.0023; branches: P=0.0066), but this was further attenuated

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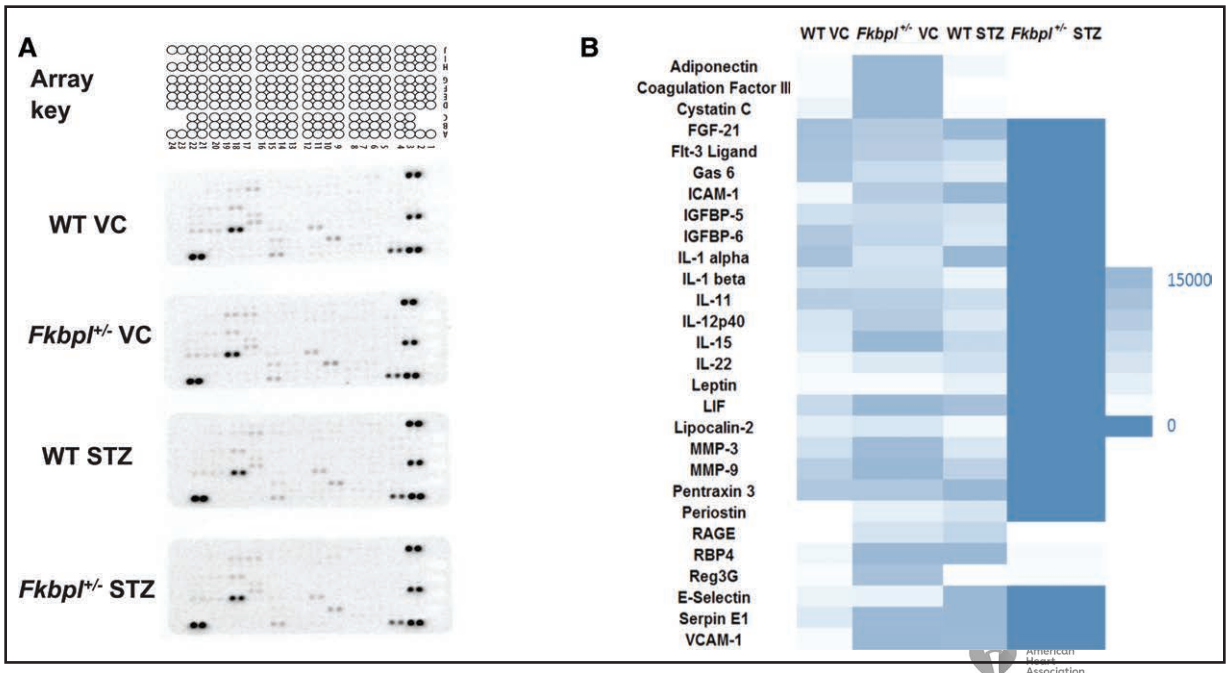

Figure 3. Myocardial inflammation is overall higher in wild-type (WT) diabetic mice and in FKBPL (FK506-binding protein–like)+/– nondiabetic mice, but it is low in *fkbpl*+/– diabetic mice.

A, Arrays of original blots. **B**, Protein abundance per group is presented as a heat map; green represents higher protein expression while red represents lower protein expression (N≥7 [biological replicates] per group, pooled). FGF indicates fibroblast growth factor; ICAM, intercellular adhesion molecule; IL, interleukin; LIF, leukemia inhibitory factor; MMP, matrix metalloproteinase; RBP, retinol-binding protein; STZ, streptozotocin; VC, vehicle control; and VCAM, vascular cell adhesion molecule.

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in high glucose (Figure 7D through 7F, n=6, junctions: $P<0.001$; branches: $P=0.0057$). These studies suggest that restoring FKBPL in high-glucose condition is not a favorable approach to stimulating angiogenesis.

MicroRNA Expression Is Differentially Regulated With FKBPL Overexpression in the Presence or Absence of High Glucose

To decipher the mechanism underlying FKBPL-mediated regulation of endothelial cell function(s) in the presence and absence of high-glucose conditions, FKBPL was overexpressed and microRNAs extracted, followed by microRNA sequencing. We performed an unsupervised analysis using multidimensional scaling to reduce the dimensionality of the data set while maintaining the distances between the elements analyzed. This approach facilitates representation of the data set in a 2-dimensional space and grouping of the samples according to their origin and treatments (Figure 8, n=3). Differentially expressed genes were then detected. In high-glucose conditions, hsa-miR-29b-3p and hsa-miR-302b-5p were significantly upregulated with FKBPL overexpression (Table 2), whereas hsa-miR-3202, hsa-miR-122-5p, and hsa_piR_020381/gb/DQ597997/Homo were significantly downregulated (Table 2). Under normal glucose conditions, hsa-miR-4706 and hsa-miR-92a-3p were significantly upregulated while hsa_piR_019521/gb/DQ596805/Homo and hsa_piR_020829/gb/DQ598677/Homo were downregulated after FKBPL overexpression (Table 3).

Pathway enrichment analysis was performed to elucidate the function of the differentially expressed microRNAs. Under high-glucose conditions, FKBPL overexpression was correlated with the upregulation of pathways involved in inflammation and vascular remodeling, including ECM (extracellular matrix) receptor interaction, TGF (transforming growth factor)-β signaling, leukocyte migration, and IL-7 signaling (Table 4).[28,29] Metabolic pathways including PI3K-Akt, insulin signaling, and glucose transport were also upregulated.[30,31] In contrast, proangiogenic signaling pathways (FGF receptor signaling, PDGF [platelet-derived growth factor] signaling and the adherens junction pathway for vascular permeability) were downregulated (Table 5), corroborating our in vitro data.

However, leukocyte activation showed contrasting trends: upregulated at high glucose (Table 4) and downregulated at normal glucose (Table 6) after FKBPL overexpression. Notably, several proinflammatory pathways TLR (toll-like receptor) signaling, MAPK (mitogen-activated protein kinase) cascade, JNK (C-jun N-terminal kinase) signaling[32–34] (Table 6) were downregulated under normal glucose conditions after FKBPL overexpression, aligned to what we observed in *fkbpl*+/– mice (Figure 3).

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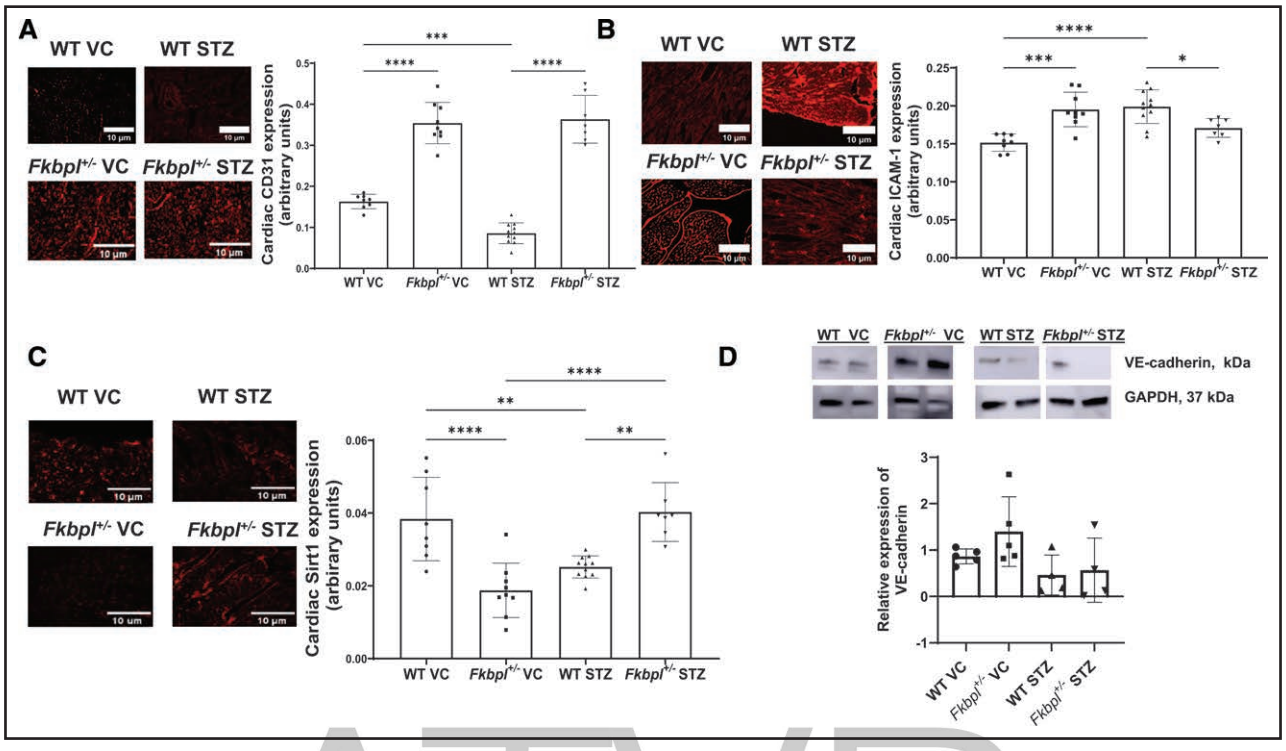

Figure 4. FKBPL (FK506-binding protein–like) deficiency modulates endothelial activation, angiogenesis, and vascular integrity in hearts of diabetic and nondiabetic mice.

Frozen left ventricle heart sections were processed and stained to quantify (**A**) CD-31, (**B**) ICAM-1 (intercellular adhesion molecule 1), and (**C**) SIRT-1 (sirtuin 1) protein expression using mean fluorescence and compared between the 4 groups; representative immunofluorescence images (red staining) included. **D**, Protein lysates were extracted from the frozen heart tissue, VE-cadherin (vascular endothelial cadherin) protein expression quantified using Western blotting relative to GAPDH expression and compared between the 4 groups; representative Western blots inset (mean±SD; n≥7 [biological replicates], 1-way ANOVA, Sidak post hoc test, *P<0.05, **P<0.01, ***P<0.001; ****P<0.0001). STZ indicates streptozotocin; VC, vehicle control; and WT, wild-type.

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FKBPL-Based Peptide Therapeutic, AD-01, Upregulates *vcam-1* and *glut1/ir* Expression in High-Glucose Conditions

To further investigate the diabetes-specific mechanisms downstream of FKBPL and explore possible targeted therapeutic options, the FKBPL-based 24-amino acid peptide mimetic, AD-01, which binds to CD44[10] but not HSP90 was used to treat HCAECs under normal and high-glucose conditions. Given, we identified in miRNAseq that miR-302b-5p was upregulated with FKBPL overexpression, in high-glucose conditions, we also investigated whether it is a critical mechanism in AD-01 treatment. Interestingly, in high-glucose conditions, AD-01 treatment at both 1 and 100 nmol/L did not affect *icam-1* gene expression (Figure 9A, n=3–6), whereas miR-302b-5p increased *icam-1* gene expression. Regarding AD-01, this is in contrast with the increased inflammatory response with FKBPL overexpression in high-glucose conditions (Table 4). On the other hand, a dose-dependent increase in *vcam-1* mRNA expression was noted after AD-01 treatment under high-glucose conditions (Figure 9B, P=0.018), with expression restored to baseline at the higher AD-01 dose (Figure 9B, P=0.042). The inhibition of miR-302b-5p did not influence 1 nmol/L AD-01-mediated increase in *vcam-1* gene expression (Figure 9B; P=0.0037). In conjunction with these changes, 1 nM AD-01 increased the expression of *cd44* in high-glucose conditions (Figure 9C, P=0.0029); no change in CD44 expression was observed with 100 nmol/L AD-01 compared with control (Figure 9C), but there was a difference between 1 nmol/L and 100 mmol/L AD-01 treatment (P=0.042). Interestingly, miR-302b-5p inhibitor did not impede the increase in cd44 with 1 nM AD-01 treatment (P=0.041). Furthermore, insulin and glucose transport pathways, which were upregulated after overexpression of FKBPL in high-glucose conditions (Table 4), showed the same pattern here with 1 nM AD-01 treatment (Figure 9D). Specifically, *glut1* and *ir* mRNA expression was upregulated by 1 nM AD-01 (Figure 9D, *glut1*: P=0.021; *ir:* P=0.025). Interestingly, 100 nmol/L AD-01 did not affect *glut1* or *ir1* expression compared with the control. The inhibition of mir302b-5p did not impact the 1 nM AD-01 upregulation of *glut1* (P=0.0035), unlike with *ir1* (P=0.68). These results suggest that AD-01 in a dose-dependent manner can increase glucose uptake by endothelial cells, which can lead to endothelial dysfunction and hence is not a good therapeutic option for early stages of vasoprotection in diabetes. MicroRNA-302b-5p is not

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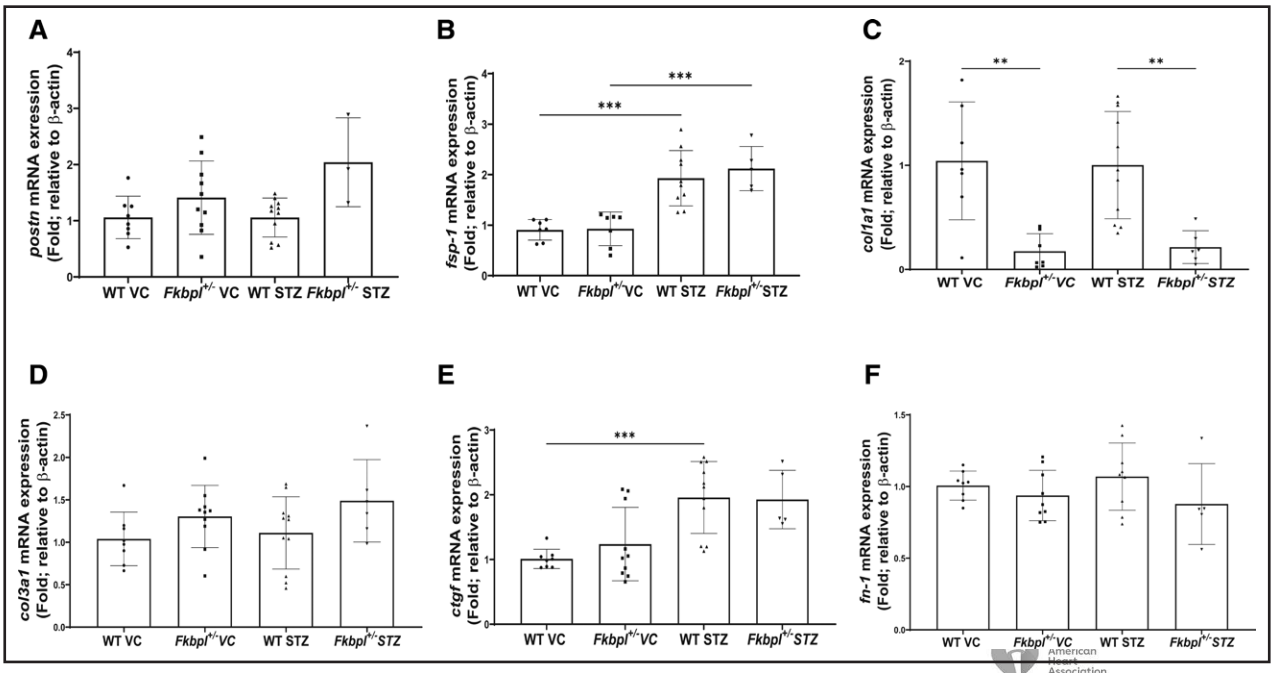

Figure 5. The expression of cardiac fibrosis gene panel in murine cardiac tissues in low FKBPL (FK506-binding protein–like) settings with or without diabetes.

Cardiac tissue was collected, RNA extracted, and quantitative polymerase chain reaction performed to quantify the mRNA expression of (**A**) *postn*, (**B**) *fsp-1*, (**C**) *col1a1*, (**D**) *col3a1*, (**E**) *ctgf*, and (**F**) *fn-1*. Mean±SD; n≥3 (biological replicates), 1-way ANOVA, Sidak post hoc test, *$P<0.05$, **$P<0.01$, ***$P<0.001$. STZ indicates streptozotocin; VC, vehicle control; and WT, wild-type.

a critical mechanism involved in AD-01–mediated effect and other compensatory mechanisms might be involved. This is aligned with our overall observation that low FKBPL expression is a protective vascular mechanism in diabetes, as shown before in other models of CVD.[16,35]

DISCUSSION

In this study, we discovered several key findings: (1) under nondiabetic conditions, low global FKBPL levels were associated with early cardiac remodeling and inflammation in vivo. (2) Remarkably, the presence of diabetes rescued these adverse features. (3) In vitro, FKBPL overexpression in endothelial cells under diabetic conditions further impaired angiogenesis, accompanied by downregulation in FGF/PDGF and upregulation in TGF-β, IL-7/leukocyte migration, and glucose/insulin transport pathways, mediated in part by differential expression of microRNAs, including upregulation of miR-29b-3p and miR-302b-5p and downregulation of miR-3202 and miR-122-5p. (4) Importantly, treatment with the FKBPL-based therapeutic peptide, AD-01, increased the expression of vascular dysfunction and glucose-uptake genes under metabolically challenged conditions, which was not completely reversed by microRNA-302b-5p, suggesting other redundant mechanisms and microRNAs are involved. These findings suggest that reduced FKBPL may have cardioprotective and vasoprotective effects in diabetes, and that its peptide derivative, AD-01, could not serve as a potential targeted therapeutic in this context. However, further in vivo studies are needed to confirm these effects.

Under Nondiabetic Conditions, Low Global FKBPL Levels Were Associated With Early Cardiac Remodeling and Inflammation

We previously reported that nondiabetic participants with CVD had higher FKBPL plasma concentration compared with those without, and FKBPL was positively correlated with biomarkers and parameters of diastolic dysfunction.[13] Consistent with this, plasma FKBPL levels are increased in HFpEF.[16] Despite these findings, the role of FKBPL in cardiac structure and function is limited. In the current study, we showed that compared with WT, *fkbpl*$^{+/-}$ mice under nondiabetic conditions exhibit altered cardiac remodeling and function, including increased cardiomyocyte area and increased E/A ratio, suggesting diastolic dysfunction and supporting our human data. Hearts from *fkbpl*$^{+/-}$ mice also exhibited reduced cardiac collagen deposition and *col1a1* mRNA. Interestingly, in our previous study using 3-dimensional bioprinted and 2-dimensional models of cardiac fibrosis, we showed that FKBPL-based peptide, AD-01, modulates *col1a1* and *mmp2* expression.[27] Although excessive fibrosis is a characteristic feature of pathological remodeling, lower basal deposition of collagen may reflect changing activity of fibroblasts or ECM turnover in the absence of diabetes. However, the parallel increment in cardiomyocyte size suggests early hypertrophic remodeling, which, over

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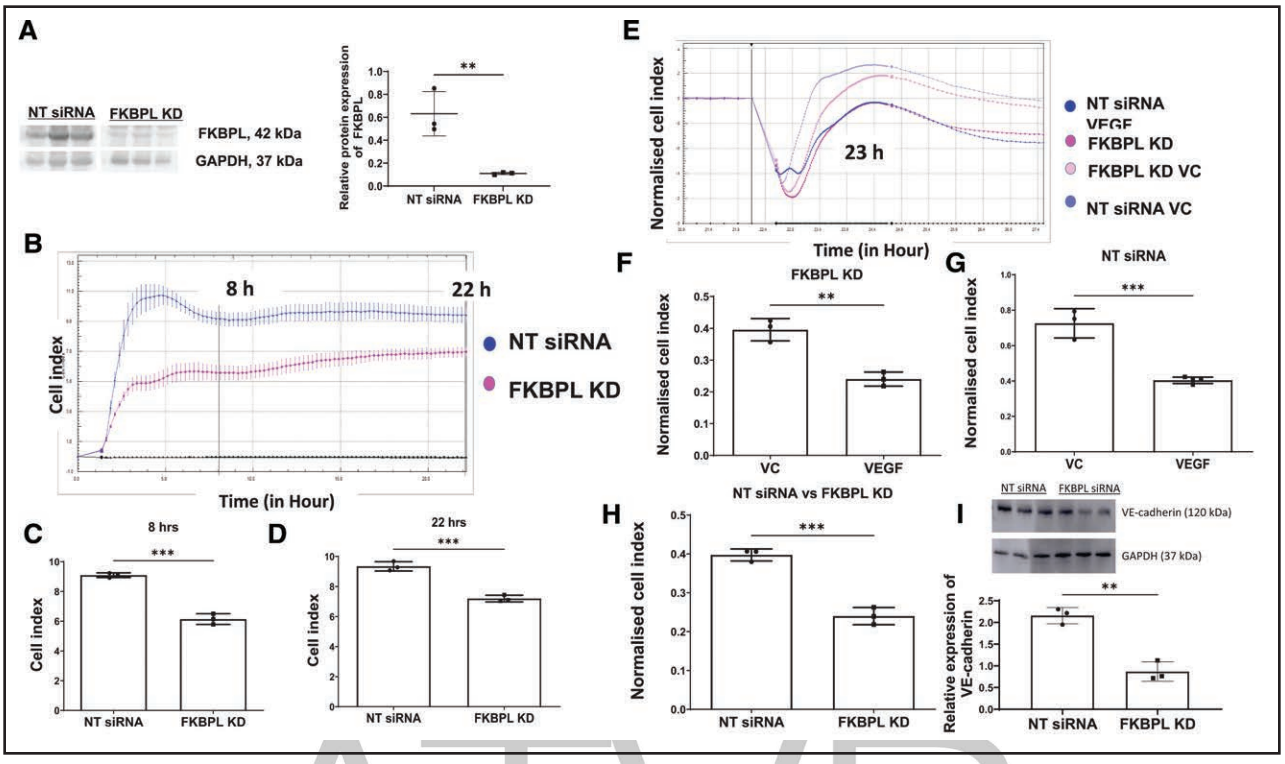

Figure 6. FKBPL (FK506-binding protein–like) silencing disrupts endothelial barrier function and augments VEGF (vascular endothelial growth factor)–induced permeability in human aortic endothelial cells (HAoECs).
A, Downregulation of FKBPL expression in HAoECs by siRNA silencing led to a significant reduction in FKBPL protein expression. **B**, FKBPL knockdown resulted in the loss of endothelial barrier integrity at (**C**) 8 hours and (**D**) 22 hours. Rapid endothelial barrier disruption was induced by (**E**) VEGF (50 ng/mL) treatment in FKBPL-silenced and control cells. VEGF reduced cell index in (**F**) FKBPL knockdown condition or (**G**) nontargeting (NT) siRNA. **H**, Loss of barrier function was markedly pronounced in the FKBPL-silenced cells. **I**, FKBPL knockdown using siRNA in HAoECs resulted in decreased VE-cadherin (vascular endothelial cadherin) protein expression (mean±SD; n=3 [biological replicates, each performed in triplicate], **<0.01, ***<0.001, unpaired Student *t* test). VC indicates vehicle control

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the longitudinal perspective, may orient to a structural or functional impairment if the compensatory mechanisms fail. Additional evaluations of myocardial stiffness, passive tension, and matrix protein turnover would help elucidate the implications of these findings.

Interestingly, these findings were associated with increased cardiac inflammatory markers, including interleukins, ICAM-1, VCAM-1, and elevated cystatin C protein expression. Cystatin C functions as a potent inhibitor of cysteine proteases. In addition to its well-recognized role as a biomarker of kidney injury, elevated cystatin C levels have been linked to CVD[36] and shown to be an independent predictor of adverse outcomes in people with chronic heart failure.[37] In contrast to our findings and these, elevated expression of Cystatin C in hearts after myocardial ischemia reperfusion injury was associated with increased collagen,[38] suggesting context specific changes. This could also be related to the duration of the experiment (ie, 12 weeks), reflective of early stages of cardiac adverse remodeling. In support of this, MMP-3/9 were also increased in non-diabetes *fkbpl*$^{+/-}$ hearts, which play an important role in the underlying pathology leading to coronary artery disease and progressive heart failure.[39] In fact, MMP-3 is a reliable predictor of cardiovascular events 5 years postcoronary artery disease diagnosis.[40]

These findings imply that in *fkbpl*$^{+/-}$ mice, cardiomyocytes undergo compensatory hypertrophy with active remodeling and inflammation, but without excessive fibrosis. These changes may reflect an early remodeling phase that could progress toward dysfunction if inflammation and matrix imbalance persist.

The in vivo results were corroborated by in vitro findings in endothelial cells, where VE-cadherin was reduced with FKBPL knockdown in endothelial cells in conjunction with impaired endothelial barrier function and reduced barrier response to stress after disruption, suggesting a critical role for FKBPL in maintaining endothelial cell function at baseline. Reduced VE-cadherin expression, an indicator of compromised endothelial barrier integrity, reflects endothelial dysfunction.[25,35]

Remarkably, the Presence of Diabetes, Rescued These Adverse Features

In contrast, under diabetic conditions in *fkbpl*$^{+/-}$ mice, the previously observed cardiac remodeling and inflammation were absent, suggesting that hyperglycemia may

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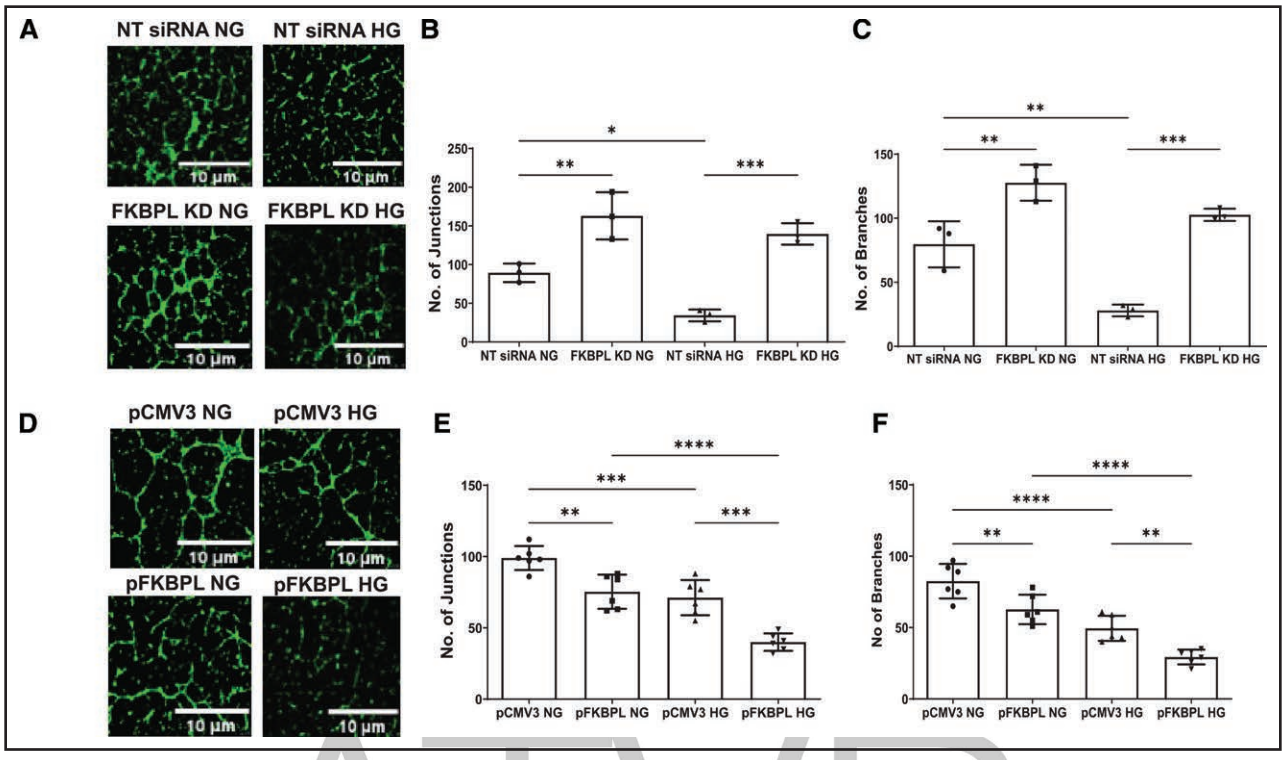

Figure 7. FKBPL (FK506-binding protein–like) plays a key role in regulating endothelial cell angiogenic potential in hyperglycemia.
A, FKBPL knockdown in human aortic endothelial cells using siRNA resulted in increased numbers of junctions (**B**) and branches (**C**) under both normal and high-glucose conditions. **D**, FKBPL overexpression using plasmid (pCMV3) inhibited junctions (**E**) and branches (**F**) formation under normal glucose and further impeded angiogenic capacity under high-glucose exposure (mean±SD; n≥3 [biological replicates, each performed in triplicates], 1-way ANOVA with Sidak post hoc test, *P<0.05, **P<0.01, ***P<0.001; ****P<0.0001). NT indicates nontargeting.

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exert a protective effect in the context of reduced FKBPL expression in the heart. This is surprising because clinical data report elevated circulating FKBPL plasma concentrations in patients with either type 2 diabetes or CVD,[13] whereas FKBPL expression is suppressed in placenta following pregnancies with T1DM.[19] Unfortunately, we could not measure plasma FKBPL concentration in these mice due to low secretion of FKBPL, which was undetectable using commercially available ELISA. This contradictory finding could reflect tissue-specific regulation mechanisms; increased circulating FKBPL may indicate systemic inflammatory or metabolic stress that originated in peripheral tissues, whereas local cardiac and endothelial FKBPL reductions may be adaptive, reflecting responses to oxidative stress and inflammation caused by hyperglycemia. However, another previous study showed that FKBPL levels are increased in preeclampsia, a CVD of pregnancy, both in plasma and in placental tissue.[41] It is possible that since our model is an early diabetes model (12 weeks of diabetes), compensatory reduction in FKBPL is not long-lasting, which is aligned with our previous observation of a positive correlation between FKBPL plasma concentration and the duration of diabetes.[13] Therefore, the FKBPL expression and regulation in different types of diabetes or CVD appears context dependent.

The protective effect of diabetes in *fkbpl*$^{+/-}$ mice may be explained by several mechanisms. First, inflammatory pathways were markedly downregulated in diabetic *fkbpl*$^{+/-}$ cardiac tissue (Figure 3). Second, we observed increased CD-31 staining in the diabetic heart, depicting increased angiogenesis as a compensatory mechanism to hyperglycemia-induced vasoconstriction.[8] As expected, the presence of diabetes led to upregulation of ICAM-1, as described previously[42]; however in low FKBPL settings, this was reversed, whereas VE-cadherin expression remained unchanged compared with diabetic WT mice. Another important vasoprotective mechanism centered around SIRT-1 was restored in diabetic *fkbpl*$^{+/-}$ mice, compared with WT diabetic mice. The activation of SIRT-1 signaling improves endothelial function by inducing eNOS (endothelial NO synthase) upregulation, thereby improving endothelial cell angiogenic ability, and reducing oxidative stress and inflammation.[43,44] Changes observed in cardiac SIRT-1 are in line with the hypothesis that FKBPL downregulation in diabetes has a vasoprotective effect and could subsequently improve NO bioavailability and suppression of cell death pathways.[45,46] Additionally, *fkbpl*$^{+/-}$ diabetic mice demonstrated a reduction in similar proinflammatory cardiac cytokines and additional few (eg, E-selectin, pentraxin 3, IGFBO5/6), implicated in the development of CVD.[47] Taken together,

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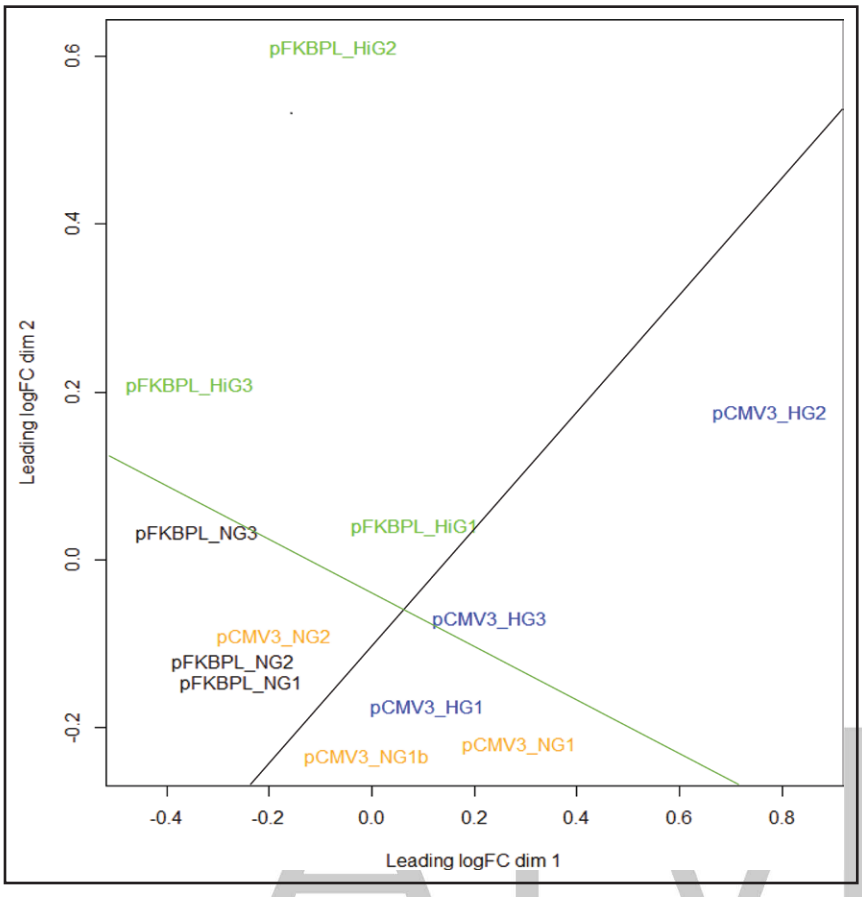

Figure 8. Multidimensional scaling for microRNA sequencing.
This analysis segregates the dimensionality of the samples analyzed to 2 dimensions (leading LogFC dim 1 and leading LogFC dim 2), representing this new reduced space. Different conditions (colored in 4 different colors) are segregated and within their respective groups in most of the conditions with the exception of samples pCMV3 NG2 and pCMV3 HG1. All other 3 conditions are clearly separated from each other. The black line represents a boundary between control samples and FKBPL (FK506-binding protein–like) overexpressed samples, and the green line represents the boundary between normal and high-glucose samples (n=3, biological replicates, each performed in triplicate [technical replicates]). pFKBPL HG denotes overexpression in high-glucose conditions; pFKBPL NG denotes overexpression in normal glucose conditions; pCMV3 HG denotes empty vector high glucose; and pCMV3 NG denotes empty vector normal glucose.

these findings suggest that low global levels of FKBPL in diabetes may exert anti-inflammatory actions, which may protect the vasculature and thereby reduce downstream cardiac remodeling and dysfunction.

Interestingly, several cytokines that were upregulated in *fkbpl*$^{+/-}$ nondiabetic mice and downregulated in *fkbpl*$^{+/-}$ diabetic mice may suggest differential effects of FKBPL on cardiac inflammatory signaling in the presence and absence of diabetes. It was interesting that diabetes induction through streptozotocin injection, although reducing FKBPL expression in WT mice by ≈30%, did not reach the same level of reduction as in *fkbpl*$^{+/-}$ mice. However, diabetes in *fkbpl*$^{+/-}$ mice seems to have further reduced FKBPL expression than diabetes in WT mice, suggesting that different mechanisms are involved as demonstrated by immunofluorescence and proteomics findings. The presence of diabetes in *fkbpl*$^{+/-}$ mice could be inducing compensatory mechanisms, including adiponectin and FGF, causing vasoprotective effect.

Table 2. List of miRNAs Which Are Significantly Influenced by FKBPL Overexpression in the Presence of High Glucose Compared With Empty Vector High-Glucose Controls

miRNA	logFC	FDR
hsa-miR-29b-3p	0.566	0.01
hsa-miR-302b-5p	0.61	0.03
hsa-miR-3202	−1.356	0.01
hsa-miR-122-5p	−0.907	0.01
hsa_piR_020381/gb/DQ597997/Homo	−1.057	0.03

FKBPL indicates FK506-binding protein–like; and miRNA, microRNA.

Endothelial Cell FKBPL Overexpression in Hyperglycemia Further Impairs Angiogenesis by Downregulating Key Proangiogenic Pathways and Upregulating Inflammation

In the presence of high glucose, we demonstrated in vitro that angiogenesis was reduced in association with FKBPL downregulation, which is likely a compensatory mechanism to reduce antiangiogenic factor production, thereby promoting angiogenesis. However, when FKBPL was overexpressed in the presence of high glucose, angiogenesis was further reduced, whereas FKBPL knockdown in high glucose resulted in increased angiogenesis. The FKBPL-mediated mechanisms were elucidated by identifying specific endothelial microRNA targets, which highlighted miR-29b-3p and miR-302b-5p as upregulated after FKBPL overexpression in high-glucose conditions. It has been previously demonstrated that miR-29b-3p expression is stimulated with LPS-induced inflammation in HAoECs and that suppression of this microRNA reduces the secretion of inflammatory

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Table 3. List of miRNAs Which Are Significantly Influenced by FKBPL Overexpression in Normal Glucose Conditions Compared With Empty Vector Normal Glucose Controls

miRNA	logFC	FDR
hsa-miR-4706	1.244	0.019
hsa-miR-92a-3p	0.444	0.032
hsa_piR_019521/gb/DQ596805/Homo	−1.302	0.0027
hsa_piR_020829/gb/DQ598677/Homo	−0.801	0.019

FKBPL indicates FK506-binding protein–like; and miRNA, microRNA.

mediators, such as TNF-α (tumor necrosis factor-α) and IL-1 (α,γ).[48] In addition, the suppression of miR-29b is reported to inhibit NF-κB (nuclear factor-κB) and JNK signaling, hence dampening inflammation.[49,50] Moreover, inhibition of miR-29b-3p was shown to attenuate atherosclerosis and inflammation in aortic tissues,[48] whereas miR-29b-3p negatively regulates PI3K/Akt and VEGF-A expression, leading to inhibition of angiogenesis.[51] In an in vitro model of spinal cord injury, miR-302b-3p played a key role in reducing inflammation by activating the PI3K/Akt pathway and suppressing NF-κB after Circ_0000962 overexpression.[52] This indicates that overexpression of FKBPL in high-glucose conditions may promote inflammation, atherosclerosis, and CVD, in diabetes via modulation of these microRNAs.[53] Indeed, upregulation of both miR-29b-3p and miR-302b-5p was found to reduce angiogenesis by targeting VEGF in endometrial carcinoma and osteosarcoma,[51,54] which is supportive of our data.

Interestingly, we also found that the expression of miR-122-5p, whose circulating levels are positively correlated with the risk of developing metabolic syndrome and type 2 diabetes, was also reduced with FKBPL overexpression in high-glucose conditions. Indeed, antagonism of miR-122-5p in mouse liver is reported to reduce plasma cholesterol levels,[55,56] suggesting that restoring FKBPL in hyperglycemia using hypoglycemic agents may positively impact glucose and fat metabolism. This should be explored in the future.

Table 4. Selection of Statistically Significant (*P*<0.05) Pathways Which Are Upregulated With FKBPL Overexpression in High-Glucose Conditions (Results Were Based on the Analysis Using mirPath v.3 on KEGG and GO Annotations)

Pathway	*P* value	Annotations	No. of genes
ECM-receptor interaction	<0.001	KEGG	15
TGF-β	<0.001	KEGG	18
Leukocyte migration	0.033	GO	15
IL-7 signaling pathway	0.039	GO	2
PI3K-Akt signaling pathway	0.029	KEGG	53
Insulin signaling pathways	0.045	KEGG	27
Glucose transport	0.042	GO	12

ECM indicates extracellular matrix; FKBPL, FK506-binding protein–like; IL, interleukin; and TGF, transforming growth factor.

Table 5. Selection of Statistically Significant (*P*<0.05) Pathways Which Are Downregulated With FKBPL Overexpression in High-Glucose Conditions (Results Were Based on the Analysis Using mirPath v.3 on KEGG and GO Annotations)

Pathway	*P* value	Annotation	No. of genes
FGF receptor signaling pathway	<0.001	GO	22
PDGF	<0.001	GO	8
Adherens junction	<0.001	KEGG	14

FGF indicates fibroblast growth factor; FKBPL, FK506-binding protein–like; and PDGF, platelet-derived growth factor.

In normal glucose conditions, different microRNAs, including miR-4706 and miR-92a-3p, were found to be upregulated with FKBPL overexpression. miR-92a-3p overexpression was previously reported to lead to reduced angiogenesis in HAoECs and in experimental hindlimb ischemia,[57] which is in line with our results and the previously published work on antiangiogenic function of FKBPL.[15] Interestingly, miR-92a is reported to be reduced in arterial blood samples from nondiabetic HF patients.[58] Indeed, our recent work showed that although FKBPL is increased in HFpEF and hypertrophic cardiomyopathy, treatment with FKBPL peptide mimetic, AD-01, seems to normalize cardiomyocyte hypertrophy through a negative feedback mechanism[16] that could involve miR-92a.

Complementary enrichment analysis showed that FKBPL overexpression in high-glucose conditions activates the ECM-receptor interaction, which stimulates GM-CSF (a member of the cytokine receptor superfamily), IL-7 signaling, and NF-κB. Notably, the latter pathway is responsible for recruiting circulating neutrophils and monocytes, thus contributing to endothelial dysfunction.[29,59–61] Indeed, leukocyte migration is also activated here, implicating the presence of inflammation leading to endothelial barrier dysfunction.[62] Similarly, adherens junction pathways were downregulated, which suggests that high levels of FKBPL in high-glucose conditions could exacerbate inflammation and endothelial barrier dysfunction. Furthermore, increased FKBPL expression in high glucose was associated with downregulation of proangiogenic FGF receptor signaling pathway and

Table 6. Selection of Statistically Significant (*P*<0.05) Pathways Which Are Significantly Downregulated With FKBPL Overexpression in Normal Glucose Conditions (Results Were Based on the Analysis Using mirPath v.3 on KEGG and GO Annotations)

Pathway	*P* value	Annotation	No. of genes
TLR signaling	<0.001	GO	16
MAPK	0.016	KEGG	34
JNK	0.020	GO	13
Leukocyte migration	<0.001	GO	16

FKBPL indicates FK506-binding protein–like; JNK, C-jun N-terminal kinase; MAPK, mitogen-activated protein kinase; and TLR, Toll-like receptors.

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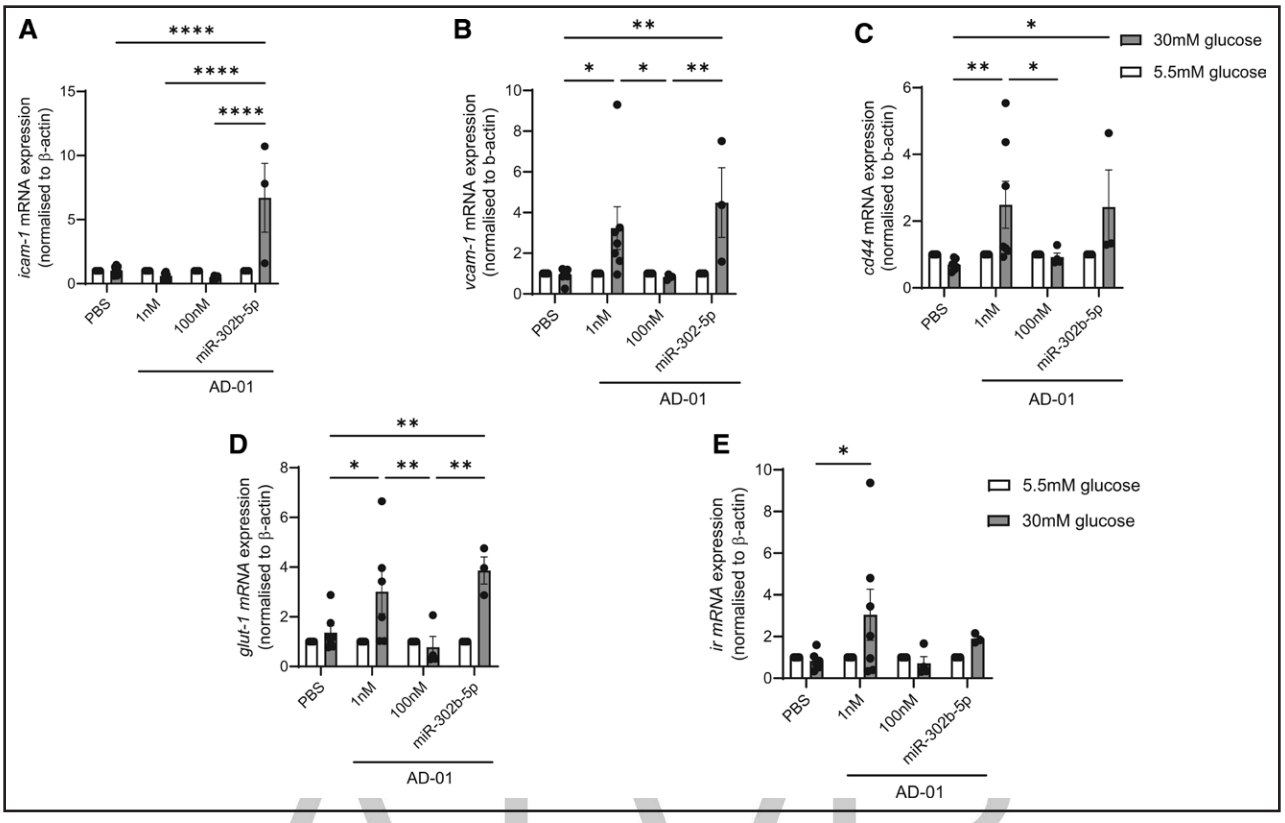

Figure 9. FKBPL (FK506-binding protein–like)-based peptide mimetic, AD-01, regulates adhesion genes, CD44, and glucose receptor.

Human coronary artery endothelial cells were treated with AD-01 (1 nM or 100 mmol/L) or microRNA-302b-5p inhibitor together with AD-01 (1 nM) in normal (5.5 mmol/L) and high-glucose (30 mmol/L) environment for 24 hours before RNA was extracted and *icam-1* (**A**), *vcam-1* (**B**), *cd44* (**C**), *glut1* (**D**), and *ir* (**E**) gene expression determined using quantitative polymerase chain reaction (mean±SEM; n=3, 2-way ANOVA with Tukey post hoc test; $^{**}P<0.01$; $^{****}P<0.0001$).

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PDGF,[63] which could worsen angiogenesis, and thereby the response of tissues to the ischemia associated with hyperglycemia. Moreover, TGF-β, with a key role in cardiac hypertrophy and fibrosis,[64] including in the presence of diabetes, was also activated with FKBPL overexpression in high-glucose conditions.

Treatment With the FKBPL-Based Therapeutic Peptide, AD-01, Increased the Expression of Vascular Dysfunction and Glucose-Uptake Genes Under Metabolically Challenged Conditions

Our study provides further mechanistic insight for the vasoprotective role of low FKBPL in diabetes and provides further evidence that using FKBPL-based peptide derivative, AD-01, is not a good therapeutic option in this context. AD-01 inhibits angiogenesis and stabilizes vasculature by binding to CD44,[10] and it is compelling that the vasoprotective and anti-inflammatory impact of low FKBPL in diabetes is mediated through a negative feedback mechanism involving the interaction of CD44 with miR-29b-3p. Previous work has shown that miR-29b-3p inhibits CD44 thus modulating immune response and inflammation.[65] Aligned to our research, a previous study has shown that miR-29b-3p was also directly linked to SIRT-1 in microvascular endothelial cells, where upregulation of miR-29b-3p led to inhibition of SIRT-1 in association with increased cell apoptosis.[66]

However, our results show that AD-01 treatment in high-glucose conditions leads to increased endothelial adhesion gene, *vcam1*, similar to what we observed in endothelial cells after overexpression of FKBPL in high-glucose conditions. This is despite an increase in proangiogenic *cd44* expression with 1 nM AD-01 dose. The dose-dependent regulation of CD44 expression has been shown before, where 1 nM AD-01 dose increased CD44 protein expression substantially >100 nM.[10] Published studies indicate that CD44 is essential for FKBPL/AD-01 activity. Specifically, FKBPL/AD-01 failed to inhibit HMEC-1 and cancer cell migration in the absence of CD44, whereas this effect was retained in CD44-expressing cells. In vitro angiogenesis was also reduced in cells with lower CD44 expression.[9,10] Furthermore, CD44, on binding to its ligand hyaluronic acid and interacting with Oct4, Sox2, and Nanog, has been shown to promote the expression of miR-302.[67] We

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have previously shown that AD-01 targets Oct4/Sox2/Nanog in cancer.[68] As discussed above, mir-302b inhibits NF-κB signaling,[52,69] a FKBPL-targeting microRNA identified in this study, which downstream inhibits adhesion molecules, including ICAM-1.[70] FKBPL and other FKBPs have already demonstrated negative regulation of NF-κB.[12,14] However, when we inhibited miR-302b-5p and added AD-01 (1 nM) treatment, this did not have any impact on AD-01-mediated upregulation of *vcam1* and *glut1* but it impeded AD-01-mediated upregulation of *ir* gene expression, suggesting that other redundant pathways might play a role, including those regulated by miR-29b-3p. This is not surprising given that FKBPL is a chaperone protein regulating multiple pathways. Taken together, our findings suggest that under high-glucose conditions, AD-01 at a 1 nmol/L dose increases CD44 expression, not impeded by inhibition of miR-302b, potentially leading to endothelial dysfunction.

Interestingly, AD-01 also upregulates *glut1* gene, responsible for glucose transport, suggesting that AD-01 could be contributing to endothelial cell dysfunction from hyperglycemia-induced damage by stimulating glucose uptake. We showed here that similar targets and pathways are targeted by FKBPL overexpression in high glucose. Nevertheless, AD-01 negative feedback mechanism on FKBPL has been previously demonstrated in 2 independent studies, in cardiomyocytes after exposure to Ang-II (angiotensin II) stress[16] and microvascular endothelial cells after exposure to hypoxia,[35] where AD-01 was capable of mitigating FKBPL expression and cellular function. For example, while hypoxia reduced the expression of FKBPL and stimulated the migration of endothelial cells, treatment with AD-01 reversed this phenotype in conjunction with normalizing FKBPL expression.[35] This was mediated via tissue remodeling and vascular integrity proteins, including collagen alpha-1(XIX) chain and JCAD (junctional cadherin–associated-5) proteins.[35] In high-glucose environment relevant to diabetes, AD-01 does not seem to have these negative feedback mechanisms, further strengthening our conclusion that low FKBPL has cardio- and vasoprotection in the context of diabetes.

The study limitations include (1) our echocardiographic assessment used E/A ratio as the main diastolic parameter. Although E/A ratio is representative of LV filling dynamics, it is known to be affected by heart rate, cardiac preload, and the depth of anesthesia[71] We did not record HR or body temperature at the time of echocardiography; however, all mice underwent the same procedures and therefore were subjected to the same effects of light anesthesia. Nevertheless, further parameters of diastolic dysfunction should be measured in the future to gain a more complete picture of the diastolic function in mice. In support of our findings, we showed previously that FKBPL is positively associated with parameters of diastolic dysfunction (LA volume, IVSd, E/A, Peak E, DT, left atrium size) in nondiabetic group of human participants.[13] Although we have now measured CD44 expression in our study, and the results support the proposed mechanism involving CD44 previously published,[9,10] this provides initial evidence for CD44's role in AD-01's mechanism of action, and direct manipulation of target microRNAs (eg, miR-302b-5p) did not seem to affect AD-01's regulation of vascular and metabolic gene expression except insulin receptor gene. Future studies should investigate other microRNAs and compensatory mechanisms to fully confirm their involvement in the context of diabetes. Future studies should test AD-01 and other treatments targeting FKBPL in vivo using nondiabetic and diabetic *fkbpl*$^{+/-}$ mice, respectively, and investigate the impact on cardiac inflammation and vascular function. As sex differences in cardiovascular and metabolic responses are well recognized, future studies should include both male and female animals to fully elucidate the role of FKBPL in diabetic cardiovascular pathology.

In conclusion, we clearly demonstrate that FKBPL downregulation under basal conditions is associated with early signs of cardiac remodeling, as well as vascular dysfunction and increased inflammation by upregulating ICAM-1, MMPs, interleukins, and downregulating SIRT-1. However, the situation appears to be reversed in diabetes, suggesting that cardiac vascular health may be improved concomitantly with reduction in inflammation. The FKBPL-mediated mechanisms in the presence of diabetes are differential involving different microRNAs (miR-302b-5p/miR-29b-3p versus miR-92a-3p) as well as inflammatory (TGF-β) and angiogenic pathways (FGF, PDGF), with some potentially compensatory mechanisms. In nondiabetic context, careful restoration of FKBPL levels toward normal physiological levels (eg, in hypoxia),[35] with AD-01, may therefore, correct angiogenic and inflammatory imbalance, restore vascular homeostasis, and prevent or delay progression of vascular dysfunction and CVD. However, in the context of diabetes, low levels of FKBPL may be protective in terms of vasculature; hence, restoring FKBPL may exacerbate associated inflammation, which could lead to accelerated vascular dysfunction and CVD. AD-01, although potentially beneficial in a nondiabetic context with low FKBPL background expression, is not suitable in the context of high glucose, even in the early stages when FKBPL is downregulated before it likely increases with longer duration of diabetes,[13] as it could exacerbate endothelial cell dysfunction from hyperglycemia-induced damage.

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Disclosures

None.

Supplemental Material

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